

TATA DATA VISUALISATION

1. Project Name

Tata Data Visualisation: Empowering Business with Effective Insights

2. Project Description

This project was completed as part of the Tata Data Visualisation Virtual Experience. The objective was to analyse an online retail transaction dataset and transform the data into clear, business-focused visual insights for senior stakeholders. The analysis focused on revenue trends, country performance, high-value customers, product demand and geographic opportunities. Microsoft Power BI was used to support data preparation, analysis, visualisation and business storytelling.

3. Business Objective

The business required an analysis of online retail transaction data to help senior stakeholders understand:

- How revenue changes over time and whether there are seasonal patterns.
- Which countries generate the highest revenue and sales volume, excluding the United Kingdom where required.
- Which customers contribute the most revenue.
- Which products and markets show strong demand.
- Which geographic areas may represent potential opportunities for further investigation.

The analysis was designed to produce clear and actionable insights rather than only present charts.

4. Data and Methodology

Dataset: Online Retail transaction data.

Data preparation:

- Reviewed transaction records for data quality.
- Removed records containing negative quantities and negative unit prices, as these were treated as invalid for the sales-performance analysis.
- Prepared the data for revenue, quantity, customer, country and time-based analysis.

Revenue approach: Revenue was analysed using transaction-level quantity and unit-price information.

Visual analysis:

- Time-series analysis for revenue trends and seasonality.
- Country-level revenue and quantity analysis.
- Customer-level revenue ranking.
- Geographic analysis to identify demand patterns.
- Power BI dashboard visualisations and interactive filtering.

Business approach: The visual findings were translated into practical recommendations for management, with emphasis on revenue growth, customer retention, inventory planning and market evaluation.

5. Key Business Questions

Q1. How does revenue trend over time, and are there significant seasonal patterns?

Q2. Which top countries generate the highest revenue, together with quantity sold, excluding the United Kingdom?

Q3. Who are the top 10 customers by revenue, and how does revenue change from the highest- to lower-value customers?

Q4. Which countries/regions show the greatest product demand and potential opportunities for business expansion, excluding the United Kingdom?

6. Key Analysis and Insights

1. Revenue Trend and Seasonality

- The first eight months show relatively stable revenue, with average monthly revenue of approximately USD 685K. Revenue begins to increase in September, rising by about 40% compared with the previous month. The upward trend continues into November, when revenue reaches approximately USD 1.5 million, the highest level observed in the available data. December data is incomplete, so no conclusion was drawn for that month.
- Overall, the pattern suggests that retail sales are strongly influenced by seasonality, particularly toward the final months of the year. This indicates that inventory planning, promotional activity and operational readiness should be considered before the seasonal peak.

2. Top Countries and Growth Opportunities

- The country analysis identified several markets with strong revenue and sales activity. The United Kingdom was excluded from the growth-opportunity visual where required because of its already high demand.
- The analysis highlighted the Netherlands, Ireland, Germany, France and Australia as markets showing significant revenue and/or sales activity in the available analysis. These markets can be considered important markets for continued customer engagement and targeted marketing.
- The analysis also identified comparatively lower-demand areas across parts of the American region, Africa, Asia and Russia. These markets should not automatically be treated as expansion opportunities; instead, they can be considered for targeted investigation using additional factors such as local demand, pricing, competition, logistics and customer behaviour.

3. Top 10 Customers by Revenue

- The customer analysis showed that revenue is distributed across the top 10 customers rather than being concentrated in only one or two customers.

- The highest-value customer generated approximately 17% more revenue than the second-highest customer. This indicates relatively low bargaining power among individual customers and a comparatively healthy distribution of revenue across high-value customers.
- From a business perspective, maintaining strong relationships with high-value customers remains important for repeat business. At the same time, the business should continue to diversify its customer base so that future revenue growth is not overly dependent on a small number of customers.

4. Product and Demand Analysis

Product demand should be evaluated alongside revenue and quantity sold to distinguish products that generate high sales value from products that move high volumes.

The Power BI analysis can be used to identify:

- Products with the highest revenue contribution.
- Products with the highest quantity sold.
- Products that show strong demand but relatively lower revenue.
- Products that may require further investigation because of comparatively low performance.

These measures provide a stronger basis for inventory planning, product prioritisation and targeted promotional activity. Product-level decisions should be based on the actual dashboard values and should be reviewed together with margin, stock availability and customer demand where those measures are available.

5. Geographic Demand and Expansion

- The geographic analysis showed that much of the strongest demand is concentrated in European markets, with the Netherlands, Ireland, Germany and France contributing significant revenue and sales activity. Australia also showed meaningful demand in the analysis.
- The analysis highlighted comparatively lower demand in parts of the American region, Africa, Asia and Russia. These areas may be considered for targeted market research rather than immediate expansion. Further validation using local market size, customer behaviour, logistics costs, pricing and competitive conditions would be appropriate before making expansion decisions.

7. Key Business Recommendations

1. Prepare inventory and marketing campaigns ahead of the strong seasonal period observed toward the end of the year.
2. Continue strengthening high-performing European markets through targeted campaigns, customer-retention activities and localised offers.
3. Maintain strong relationships with the top revenue-generating customers while continuing to diversify the customer base.
4. Monitor product-level revenue and quantity together so that inventory and promotional decisions are based on both value and demand.

5. Investigate lower-performing geographic markets before expansion, using localised pricing, product demand, logistics feasibility and customer research.
6. Use the Power BI dashboard as a recurring management tool to monitor revenue, customer value, product demand and regional performance.
7. Revisit the analysis when complete-period data becomes available, especially for December, before drawing conclusions about the full-year seasonal pattern.

8. Business Impact

- The analysis demonstrates how transactional retail data can be converted into business-focused insights. The findings support practical decisions around seasonal planning, customer retention, product monitoring and geographic market evaluation.
- The combination of data preparation, Power BI visualisation, quantitative analysis and business recommendations provides a structured approach for management reporting and recurring performance monitoring.

9. Technical Skills Demonstrated

- Microsoft Power BI
- Data cleaning and preparation
- Revenue and quantity analysis
- Time-series analysis
- Country and geographic analysis
- Customer segmentation/ranking
- Product-demand analysis
- Data visualisation and dashboard interpretation
- Business insight generation
- Stakeholder-focused reporting
- Data storytelling

10. Project Files

The GitHub repository contains the project documentation and supporting files, including the Power BI analysis file, final project report, certificate and README documentation.

Repository: https://github.com/Mahanand4/Tata_Data_Visualisation

11. Project Outcome

- The project demonstrates the ability to translate transactional retail data into understandable visual insights and business recommendations. It addresses the requirements of the Tata virtual experience by combining data preparation, visual analysis, stakeholder-focused questions and actionable conclusions.
- The project also demonstrates the use of Power BI to communicate trends and performance patterns in a way that can support management decision-making.

12. Submitted By

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